

# CS3002E Compiler Design

Monsoon 2025

Tutorial #3

Predictive Parsing- Computing FIRST and FOLLOW

Department of Computer Science & Engineering  
NIT Calicut

# Computing FIRST<sup>1</sup>

1. For a terminal  $a$ ,  $FIRST(a) = \{a\}$ .
2. If  $X \rightarrow \epsilon$ , add  $\epsilon$  to  $FIRST(X)$ .
3. For a string  $\alpha$ ,
  - 3.1 If  $\alpha$  is of the form  $a\alpha'$ , then  $FIRST(\alpha) = \{a\}$ .
  - 3.2 If  $\alpha$  is of the form  $X_1X_2..X_k$ ,  $k \geq 1$ ,
    - 3.2.1 add everything in  $FIRST(X_1)$ , except  $\epsilon$  to  $FIRST(\alpha)$ .
    - 3.2.2 if for some  $i$ ,  $FIRST(X_1X_2..X_i)$  contains  $\epsilon$ , add everything in  $FIRST(X_{i+1})$ , except  $\epsilon$  to  $FIRST(\alpha)$ .
    - 3.2.3 if for all  $i = 1 \dots k$ ,  $FIRST(X_i)$  contains  $\epsilon$ , then add  $\epsilon$  to  $FIRST(\alpha)$ .
4. If  $A \rightarrow \alpha$ , add everything in  $FIRST(\alpha)$  to  $FIRST(A)$ .

---

<sup>1</sup>Refer ALSU 4.4.2: Algorithm to compute FIRST(X) for all grammar symbols X

Problem 1: For each production in the below given grammars, compute FIRST sets for its body and head.

1.  $S \rightarrow AB$

$$A \rightarrow aA \mid \epsilon$$

$$B \rightarrow bB \mid \epsilon$$

2.  $S \rightarrow (AB)$

$$A \rightarrow b \mid \epsilon$$

$$B \rightarrow c \mid \epsilon$$

3.  $S \rightarrow AB$

$$A \rightarrow b \mid \epsilon$$

$$B \rightarrow c \mid \epsilon$$

4.  $E \rightarrow TE'$

$$E' \rightarrow +TE' \mid \epsilon$$

$$T \rightarrow \mathbf{id}$$

# Computing FOLLOW

**Case 1: Place \$ in FOLLOW(S), where S is the start symbol.**

Example:

$$S \rightarrow bAc$$

$$A \rightarrow \mathbf{id} \mid \mathbf{num} \mid \epsilon$$

$$S\$ \Rightarrow bAc\$ \Rightarrow b \mathbf{id} c\$$$

$$S\$ \Rightarrow bAc\$ \Rightarrow b \mathbf{num} c\$$$

$$S\$ \Rightarrow bAc\$ \Rightarrow bc\$$$

# Computing FOLLOW

**Case 2:** If  $A \rightarrow \alpha B \beta$ , add everything in  $FIRST(\beta)$  except  $\epsilon$  to  $FOLLOW(B)$ .

Example:

$$S \rightarrow bAc$$

$$A \rightarrow \mathbf{id} \mid \mathbf{num} \mid \epsilon$$

- Because of  $S \rightarrow bAc$ ,  $c \in FOLLOW(A)$   
 $S\$ \Rightarrow bAc\$ \Rightarrow bc\$$
- At node  $A$ , parser should choose  $A \rightarrow \epsilon$ , when lookahead is  $c$

# Computing FOLLOW

**Case 3: If  $A \rightarrow \alpha B$ , add everything in  $FOLLOW(A)$  to  $FOLLOW(B)$ .**

Example:

$S \rightarrow (A)$

$A \rightarrow \text{id}B$

$B \rightarrow +\text{id} \mid \epsilon$

Since  $A \rightarrow \text{id}B$ , everything in  $FOLLOW(A)$  is in  $FOLLOW(B)$ .

$S\$ \Rightarrow (A)\$ \Rightarrow (\text{id}B)\$ \Rightarrow (\text{id})\$$   
 $) \in FOLLOW(B)$ , because of the sentential form  $(\text{id}B)$ .

At node  $B$ , choose  $B \rightarrow \epsilon$  when the lookahead is  $)$

# Computing FOLLOW: Example

$S \rightarrow \mathbf{idA}$

$A \rightarrow +\mathbf{id} \mid \epsilon$

$S\$ \Rightarrow \mathbf{idA\$} \Rightarrow \mathbf{id\$}$

- $\$ \in FOLLOW(S)$ , and hence  $\$ \in FOLLOW(A)$
- At node A, Parser should choose  $A \rightarrow \epsilon$  when lookahead is \$.

# Computing FOLLOW

**Case 4: If there is a production  $A \rightarrow \alpha B \beta$ , such that  $FIRST(\beta)$  contains  $\epsilon$ , add everything in  $FOLLOW(A)$  to  $FOLLOW(B)$ .**

Example:

$$E \rightarrow TE'$$

$$E' \rightarrow +TE' \mid \epsilon$$

$$T \rightarrow \mathbf{id}$$

Since  $E \rightarrow TE'$ , and  $FIRST(E')$  contains  $\epsilon$ , add everything in  $FOLLOW(E)$  to  $FOLLOW(T)$ .

$$E\$ \Rightarrow TE'\$ \Rightarrow T\$ \Rightarrow \mathbf{id\$}$$



# Computing FOLLOW

To compute  $FOLLOW(A)$  for all nonterminals  $A$ , apply the following rules until nothing more can be added to any of the  $FOLLOW$  sets:

1. Place  $\$$  in  $FOLLOW(S)$  where  $S$  is the start symbol.
2. If there is a production  $A \rightarrow \alpha B \beta$ , add everything in  $FIRST(\beta)$ , except  $\epsilon$  to  $FOLLOW(B)$ .
3. If there is a production  $A \rightarrow \alpha B$ , add everything in  $FOLLOW(A)$  to  $FOLLOW(B)$
4. If there is a production  $A \rightarrow \alpha B \beta$ , such that  $FIRST(\beta)$  contains  $\epsilon$ , add everything in  $FOLLOW(A)$  to  $FOLLOW(B)$ .

## Problem 2: Compute FOLLOW for each nonterminal

1.  $S \rightarrow AB$   
 $A \rightarrow aA \mid \epsilon$   
 $B \rightarrow bB \mid \epsilon$

2.  $S \rightarrow (A)$   
 $A \rightarrow BC$   
 $B \rightarrow b \mid \epsilon$   
 $C \rightarrow c \mid \epsilon$

3.  $A \rightarrow BC$   
 $B \rightarrow b \mid \epsilon$   
 $C \rightarrow c \mid \epsilon$

# Predictive Parsing Table Construction

For each production  $A \rightarrow \alpha$

1. for each terminal  $a \in FIRST(\alpha)$ , add  $A \rightarrow \alpha$  to  $T[A, a]$ .
2. if  $\epsilon$  is in  $FIRST(\alpha)$ , for each terminal  $b$  in  $FOLLOW(A)$ , add  $A \rightarrow \alpha$  to  $T[A, b]$ .
3. if  $\epsilon$  is in  $FIRST(\alpha)$ , and  $\$$  is in  $FOLLOW(A)$ , add  $A \rightarrow \alpha$  to  $T[A, \$]$ .

# Predictive Parsing Table: example

$$E \rightarrow TE'$$

$$E' \rightarrow +TE' \mid \epsilon$$

$$T \rightarrow id$$

|    | id                  | +                     | \$                        |
|----|---------------------|-----------------------|---------------------------|
| E  | $E \rightarrow TE'$ |                       |                           |
| T  | $T \rightarrow id$  |                       |                           |
| E' |                     | $E' \rightarrow +TE'$ | $E' \rightarrow \epsilon$ |

## Problem 3: Build Predictive Parsing Table for each grammar

1.  $S \rightarrow AB$   
 $A \rightarrow aA \mid \epsilon$   
 $B \rightarrow bB \mid \epsilon$

2.  $S \rightarrow (A)$   
 $A \rightarrow BC$   
 $B \rightarrow b \mid \epsilon$   
 $C \rightarrow c \mid \epsilon$

3.  $A \rightarrow BC$   
 $B \rightarrow b \mid \epsilon$   
 $C \rightarrow c \mid \epsilon$

## Reference:

- Aho A.V., Lam M.S., Sethi R., and Ullman J.D. Compilers: Principles, Techniques, and Tools (ALSU). Pearson Education, 2007.

## Further reading:

- ALSU chapter 4, sections 4.4.2, 4.4.4